

OptoLink: Breaking Memory Bandwidth Bottlenecks in FHE Accelerators with Photonic Interconnects

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Fully Homomorphic Encryption (FHE) enables secure computation directly over encrypted data, making it highly valuable for domains such as healthcare, finance, and cloud services. However, its deployment is still constrained by immense computational overheads and critical memory bandwidth limitations, particularly in CKKS bootstrapping and the Number Theoretic Transform (NTT). While recent hardware accelerators have improved arithmetic throughput, they remain constrained by inefficient memory transfers and off-chip communication overhead. To overcome these limitations, we introduce *OptoLink*, a photonic interconnect architecture at the chiplet scale. By exploiting Wavelength Division Multiplexing (WDM) and Space Division Multiplexing (SDM), *OptoLink* delivers ultra-high throughput and low-latency data transfer. Our design achieves up to 1.6 TB/s of bandwidth across 128 optical channels, representing a 300× latency reduction compared with conventional electronic network. In addition, *OptoLink* provides efficient broadcast and multicast capabilities, significantly reducing redundant data movement. Using an extended FHE simulation framework, we show that *OptoLink* improves CKKS bootstrapping throughput by up to 11× on HEAX and 1.6× on F1 and ARK, while reducing memory transfer delays by orders of magnitude. Furthermore, encrypted machine learning workloads, including logistic regression training and ResNet-20 inference, benefit from higher throughput and alleviated bandwidth pressure, demonstrating the potential of *OptoLink* to enable practical large-scale FHE acceleration.

CCS Concepts: • **Hardware** → **Photonic and optical interconnect**; *Emerging architectures*; • **Security and privacy** → *Cryptography*.

Additional Key Words and Phrases: Fully Homomorphic Encryption (FHE), Photonic Interconnects, Wavelength Division Multiplexing, Memory Bandwidth, 2.5D Integration, NTT Acceleration, Post-Quantum Cryptography

1 Introduction

Fully Homomorphic Encryption (FHE) has emerged as a cornerstone of privacy-preserving computation, allowing operations to be carried out directly on encrypted data without exposing the underlying plaintext. This property is highly attractive in domains such as cloud computing, healthcare, and finance, where sensitive information must remain secure throughout computation [1, 13, 18]. As shown in Fig. 1, a typical FHE workflow involves the client encrypting data locally before outsourcing computation to an untrusted server, which operates solely on ciphertexts and returns encrypted results. Since the secret key never leaves the client, strong security guarantees are maintained even in adversarial settings. Moreover, FHE schemes are built on lattice-based hardness assumptions and are believed to be secure against quantum adversaries, making them strong candidates

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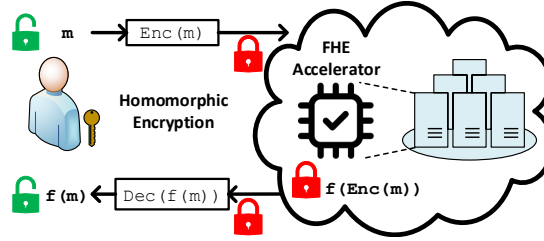


Fig. 1. End-to-end workflow of fully homomorphic encryption (FHE), where encrypted data is processed by an accelerator without exposing the underlying plaintext.

in the post-quantum cryptography landscape [58]. Despite its promise, the practical deployment of FHE remains limited by substantial computational overheads. Operations on ciphertexts are often several orders of magnitude slower than their plaintext counterparts [33], largely due to noise terms that accumulate with each homomorphic operation. Once this noise exceeds a threshold, decryption fails, making noise control a central challenge in all FHE schemes.

Among existing schemes, Cheon–Kim–Kim–Song (CKKS) [17] is widely used for approximate arithmetic, particularly in machine learning (ML) and analytics. However, CKKS suffers from rapid noise growth under multiplications. To address this challenge, CKKS employs *bootstrapping*, a computationally intensive re-encryption procedure that refreshes ciphertexts to reduce accumulated noise, but it can be up to two orders of magnitude slower than basic addition or multiplication operations [32]. In practice, bootstrapping dominates runtime, often accounting for up to 80% of execution time [35, 36, 51]. Moreover, it requires gigabytes of working memory, forcing frequent off-chip data transfers [35, 36, 51, 52]. As a result, performance is overwhelmingly constrained by memory bandwidth.

A second performance-critical kernel is the *Number Theoretic Transform (NTT)*, the modular analogue of FFT used in polynomial multiplication. NTT reduces asymptotic complexity from $O(n^2)$ to $O(n \log n)$ [25, 41], yet still dominates execution cost, often contributing over half of ciphertext multiplication and encrypted inference time [47, 49]. Its large bandwidth demands and frequent memory-access conflicts limit scalability [48, 60], especially at high-security levels where polynomial sizes and twiddle tables expand significantly [61].

Challenge. Numerous hardware accelerators have sought to mitigate these costs. For example, F1 introduces explicit memory hierarchies to improve ciphertext reuse [51], CraterLake deploys vector uniprocessors with custom functional units to reduce bandwidth needs [52], and FPGA-based designs such as FAB exploit datapath optimizations and subroutine fusion [36]. While effective, these solutions still require gigabytes of off-chip data movement, leaving performance bottlenecked by memory bandwidth. NTT is a principal driver of this demand, as its repeated coefficient and twiddle-factor fetches create severe bandwidth pressure and access conflicts [48, 60]. Pipeline optimizations [43] provide partial relief but often reduce efficiency. These limitations underscore the urgent need for new interconnect and memory technologies to enable scalable FHE acceleration.

Proposal. We advocate the integration of photonic interconnects into FHE accelerator architectures as a natural solution to these challenges. Optical links replace traditional electrical wiring with high-bandwidth, low-latency communication channels that simplify multiplexer structures in NTT designs [48], alleviate memory contention, and sustain multi-gigabyte streaming demands of CKKS bootstrapping. Furthermore, photonics natively support efficient *one-to-many* communication via wavelength- and space-division multiplexing [7], enabling scalable multi-chiplet systems. While such benefits have been demonstrated in deep neural network accelerators [39], and our earlier study have explored their feasibility for FHE workloads [50], their full potential for scalable FHE acceleration remains largely unexplored.

This paper introduces *OptoLink*, a chiplet-based FHE accelerator architecture that leverages photonic interconnects to address bandwidth and memory challenges. Our contributions are as follows:

- (1) Unlike existing works [60] that focuses on compute acceleration for FHE, we formalize the limitations as a memory bandwidth bottleneck by investigating the requirements of state-of-the-art FHE accelerators. *Our investigation concludes that it is unrealistic to achieve practical FHE with compute acceleration alone.*
- (2) We propose *OptoLink*, an FHE-aware photonic interconnect that exploits the structured data reuse patterns of CKKS primitives to dynamically support unicast, broadcast, and multicast communication, thereby mitigating memory access conflicts and providing scalable, high-bandwidth data delivery.
- (3) We design and evaluate a scalable *OptoLink* system using photonic process design kits (PDKs) and electronic-photonic design automation (EPDA) tools, demonstrating up to 1.6 TB/s bandwidth with 128 optical channels and potential for further scaling.
- (4) Using a custom framework, we benchmark CKKS workloads ranging from primitive operations to end-to-end applications such as ResNet-20 inference [37] and logistic regression [29], demonstrating that *OptoLink* achieves up to 300× lower memory access latency and higher throughput compared to conventional electrical interconnects.

The rest of the paper is organized as follows: Sec. 2 presents the background and limitations of existing work. While Sec. 3 describes the motivation behind *OptoLink* and details the methodology, including the specific design and implementation of *OptoLink* system optimized for FHE accelerators. Sec. 4 presents the results and analysis before concluding the paper in Sec. 7.

2 Background and Motivation

This section provides an overview of FHE fundamentals, the CKKS scheme, the NTT operation, and the key limitations of current FHE accelerators.

2.1 Homomorphic Encryption (HE)

Homomorphic encryption (HE) is a class of cryptographic techniques that enable computations to be carried out directly on encrypted data, producing encrypted results that, once decrypted, are identical to the outcome of the same operations performed on the plaintexts. This property makes HE highly suitable for privacy-preserving computation scenarios, such as secure outsourcing of data processing to untrusted cloud environments.

Two major categories of HE are commonly distinguished: **Leveled Homomorphic Encryption (LHE)** and **Fully Homomorphic Encryption (FHE)**. LHE supports only a limited number of operations on ciphertexts due to the progressive accumulation of noise, making it suitable primarily for simple ML models with shallow architectures. For instance, LoLa achieves efficient inference under LHE but at the expense of reduced accuracy [12]. In contrast, deeper neural networks such as ResNet-20 [37] require substantially more operations, which necessitate FHE. FHE addresses noise growth through a bootstrapping mechanism that refreshes ciphertexts, enabling unlimited operations without compromising correctness [28].

Different FHE schemes are tailored to specific data types. For example, BGV [10] and TFHE [19] efficiently support integer and boolean computations, respectively. In contrast, the **CKKS scheme** is designed to handle approximate arithmetic over fixed-point complex numbers [17]. This property is particularly important for ML applications, where real-valued computations dominate. By encrypting approximate numerical values, CKKS trades exactness for efficiency and has become a widely adopted solution in ML as a service (MLaaS) and other real-world applications.

From a security perspective, most modern FHE schemes, including CKKS, are built on the hardness of the Ring Learning With Errors (RLWE) problem. This problem is believed to remain secure even in the presence of quantum adversaries [5]. As such, FHE constitutes a key building block in post-quantum cryptography, offering both practicality and long-term security for privacy-preserving computation.

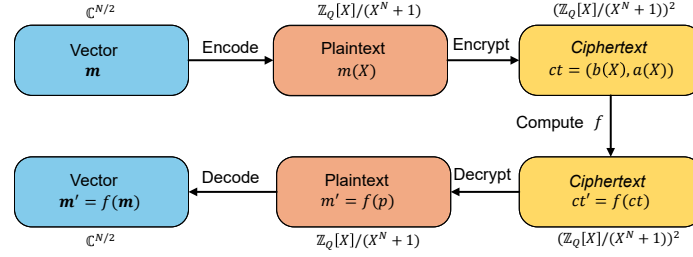


Fig. 2. Overview of the CKKS homomorphic encryption workflow, illustrating the transformation of input data through encoding, encryption, computation on ciphertexts, and decryption.

2.2 CKKS Scheme and Operations

Among existing FHE schemes, the CKKS construction [17] has gained prominence because it supports approximate arithmetic over vectors of complex numbers. Unlike exact homomorphic schemes, CKKS is designed to efficiently perform computations on real or complex floating-point values, making it highly relevant for applications in ML, large-scale data analytics, and scientific computing.

A plaintext in CKKS is represented as a vector in $\mathbb{C}^n$, where $n = N/2$ and N is a power of two corresponding to the polynomial modulus degree. Messages are encoded into polynomials within the cyclotomic ring,

$$R_Q = \mathbb{Z}_Q[X]/(X^N + 1), \quad (1)$$

with coefficients reduced modulo the ciphertext modulus Q . Through this encoding, multiple complex values can be packed into a single plaintext polynomial, enabling efficient SIMD-style (single instruction, multiple data) operations.

As illustrated in Fig. 2, the CKKS workflow begins with encoding a complex vector $m \in \mathbb{C}^{N/2}$ into a plaintext polynomial $m(X) \in R_Q$. This plaintext is then encrypted under a secret key $s(X) \in R_Q$ by sampling a random polynomial $a(X)$ and a small Gaussian error $e(X)$, resulting in the ciphertext $ct = (b(X), a(X)) = (a(X) \cdot s(X) + m(X) + e(X), a(X))$, where $ct \in (R_Q)^2$.

Decryption proceeds by computing $m'(X) = b(X) - a(X) \cdot s(X) = m(X) + e(X)$ and then decoding back into an approximate complex vector $m' \approx m$. The decryption error is inherent to CKKS but remains acceptable for applications tolerant of small numerical inaccuracies.

CKKS supports several primitive operations that act on ciphertexts while preserving their semantic correspondence to operations on plaintexts. Let $\llbracket \mathbf{m} \rrbracket$ denote the ciphertext of a vector $\mathbf{m} \in \mathbb{C}^n$:

- **Addition:** For ciphertexts $\llbracket \mathbf{m}_1 \rrbracket$ and $\llbracket \mathbf{m}_2 \rrbracket$, the addition operation is

$$\llbracket \mathbf{m}_1 \rrbracket + \llbracket \mathbf{m}_2 \rrbracket \mapsto \llbracket \mathbf{m}_1 + \mathbf{m}_2 \rrbracket, \quad (2)$$

corresponding to slot-wise addition of the plaintext vectors.

- **Multiplication:** Given $\llbracket \mathbf{m}_1 \rrbracket$ and $\llbracket \mathbf{m}_2 \rrbracket$, multiplication yields

$$\llbracket \mathbf{m}_1 \rrbracket \cdot \llbracket \mathbf{m}_2 \rrbracket \mapsto \llbracket \mathbf{m}_1 \odot \mathbf{m}_2 \rrbracket, \quad (3)$$

where $\odot$ denotes element-wise multiplication across slots.

- **Rotation:** For a ciphertext $\llbracket \mathbf{m} \rrbracket$ and integer offset k , a cyclic slot rotation is defined as

$$\llbracket \mathbf{m} \rrbracket \mapsto \llbracket \phi_k(\mathbf{m}) \rrbracket, \quad (4)$$

where ϕ_k shifts the vector entries by k positions. For instance, if $\mathbf{x} = (x_0, x_1, \dots, x_{n-2}, x_{n-1})$, then $\phi_1(\mathbf{x}) = (x_{n-1}, x_0, \dots, x_{n-3}, x_{n-2})$.

- **Conjugation:** CKKS also supports complex conjugation on encrypted slots. For a ciphertext $\llbracket \mathbf{m} \rrbracket$, the operation is

$$\llbracket \mathbf{m} \rrbracket \mapsto \llbracket \overline{\mathbf{m}} \rrbracket, \quad (5)$$

where $\overline{\mathbf{m}} = (\overline{m}_0, \overline{m}_1, \dots, \overline{m}_{n-1})$ denotes component-wise conjugation of the underlying plaintext vector.

These fundamental operations enable more complex transformations, such as matrix–vector multiplications and convolutions, which underpin privacy-preserving ML and data analysis.

2.3 Key-Switching in CKKS

Certain homomorphic operations in CKKS, such as multiplication, rotation, and conjugation, alter the ciphertext structure in a way that makes it decryptable only under a modified secret key. To restore compatibility with the original secret key, a procedure known as **key-switching** is applied [11].

Key-switching relies on specially generated public keys, called *evaluation keys*, which are derived from the original secret key. Given a ciphertext $\llbracket m \rrbracket_s$ encrypted under a secret key s , and a switching key $\text{sk}_{s \rightarrow s'}$, the operation produces a new ciphertext $\llbracket m \rrbracket_{s'}$ that encrypts the same plaintext but is decryptable under s' [11]. In CKKS, evaluation keys are instantiated for specific operations: multiplication uses evk_{mult} , while each rotation offset r requires a corresponding rotation key $\text{evk}_{\text{rot}}^{(r)}$ [32]. At a high level, the subroutine decomposes the ciphertext into smaller components using RNS basis decomposition, followed by a *modulus extension* step to align with the evaluation key domain. The resulting inner products with the evaluation key allow re-encryption under the desired secret key [30]. Since these steps involve multiple NTTs and basis conversions, key-switching is among the most computationally expensive primitives in homomorphic encryption and often dominates runtime in practical applications [32]. Importantly, key-switching exhibits substantial reuse of read-only data. Evaluation keys are fixed for a given secret key and operation and are repeatedly consumed across multiple key-switching invocations. Moreover, within a single key-switching execution, the same evaluation-key components are reused across parallel RNS limbs, digit decompositions, and NTT stages, while the ciphertext fragments themselves are distinct and naturally partitioned. This asymmetric access pattern, in which large evaluation key blocks are reused by many parallel compute units, creates inherent one to many communication opportunities.

Despite its cost, key-switching is indispensable for enabling advanced operations in CKKS, ensuring that ciphertexts remain usable across successive transformations while maintaining correctness and security.

2.4 Bootstrapping in CKKS

One of the central challenges in FHE is the continuous growth of noise and the reduction of the ciphertext modulus as multiplications are performed. When the modulus becomes too small, further operations cannot be executed without compromising correctness. To overcome this limitation, CKKS introduces bootstrapping, a refreshing procedure that restores the modulus of a ciphertext while keeping accumulated noise within acceptable bounds. This process effectively replenishes the *multiplicative depth*, enabling the scheme to support an unlimited number of computations and thereby achieving full homomorphism.

The general structure of CKKS bootstrapping has remained consistent across recent advancements [9, 14, 16]. The procedure begins with a *ModUp* operation, which increases the modulus of the ciphertext. The main difficulty lies in then homomorphically carrying out a modular reduction, which is achieved through polynomial approximation techniques. Since these approximations are numerical rather than exact, CKKS bootstrapping is inherently approximate, yet still sufficiently precise for real-number applications.

The bootstrapping workflow can be summarized in three stages:

- (1) **Forward linear transform (CoeffToSlot):** The ciphertext is first transformed from its slot representation into a coefficient representation using a homomorphic Discrete Fourier Transform (DFT). This step, carried out through repeated plaintext matrix–vector multiplications, prepares the data for modular reduction. The

number of iterations is governed by the parameter `fftIter`, which trades off matrix dimension against the total number of multiplications.

- (2) **Polynomial modular reduction:** In the coefficient domain, an approximation of the modular reduction function is homomorphically evaluated using polynomial interpolation, often combined with auxiliary exponentiation steps. The quality of this approximation directly impacts both the precision of the refreshed ciphertext and the computational efficiency of bootstrapping.
- (3) **Backward linear transform (SlotToCoeff):** After reduction, the ciphertext is mapped back to its slot representation through an inverse homomorphic DFT. Like the forward phase, this relies on multiple matrix–vector multiplications and restores the ciphertext to a form suitable for continued encrypted computation.

In essence, CKKS bootstrapping ensures that ciphertexts can be refreshed and reused indefinitely, albeit at the cost of a computationally demanding procedure. Despite ongoing advances in polynomial approximation methods, Fourier-based transforms, and optimized implementations that help reduce its computational burden, bootstrapping remains highly memory-intensive, requiring frequent access to large auxiliary key sets and polynomial data. A significant portion of this cost stems from the heavy reliance on the NTT, which underpins polynomial arithmetic in CKKS and is repeatedly invoked during key-switching, modulus conversion, and bootstrapping.

2.5 Number Theoretic Transform (NTT)

Efficient polynomial arithmetic is at the core of FHE, particularly in the bootstrapping procedure described above. The NTT provides a specialized analogue of the Fast Fourier Transform (FFT), adapted to modular arithmetic over finite fields. Unlike FFT, which operates on complex numbers, NTT works entirely over integers, making it well-suited for lattice-based cryptography and exact modular computations.

The transform is defined over the polynomial ring in Eq. 1 where q is a prime modulus chosen such that $q \equiv 1 \pmod{n}$. This condition guarantees the existence of a primitive n -th root of unity ω , satisfying $\omega^n \equiv 1 \pmod{q}$. Given a polynomial $a(x) = \sum_{j=0}^{n-1} a_j x^j$, its NTT representation is computed as

$$\tilde{a}_i = \sum_{j=0}^{n-1} a_j \omega^{i \cdot j} \pmod{q}, \quad i = 0, 1, \dots, n-1, \quad (6)$$

where $\omega^{i \cdot j}$ are the *twiddle factors*. These factors allow convolution operations in the coefficient domain to be expressed as element-wise multiplications in the NTT domain.

Thus, polynomial multiplication $c(x) = a(x) \cdot b(x)$ can be carried out efficiently as

$$c = \text{INTT}(\text{NTT}(a) \circ \text{NTT}(b)), \quad (7)$$

where $\circ$ denotes pointwise multiplication. The inverse transform restores the coefficients of the product:

$$a_j = \frac{1}{n} \sum_{i=0}^{n-1} \tilde{a}_i \cdot \omega^{-i \cdot j} \pmod{q}, \quad (8)$$

with normalization handled by the modular inverse of n .

The efficiency of NTT-based multiplication is further improved using *butterfly algorithms*, such as Cooley–Tukey [22] and Gentleman–Sande [27], which recursively divide the problem size, reducing polynomial multiplication complexity from $O(n^2)$ to $O(n \log n)$. In the Cooley–Tukey formulation, the NTT is evaluated in $\log_2 n$ stages, where each stage consists of a set of butterfly operations. A key property of this algorithm is the *reuse of twiddle factors within each stage*. Specifically, at stage s , only 2^s distinct twiddle factors are required, and each is

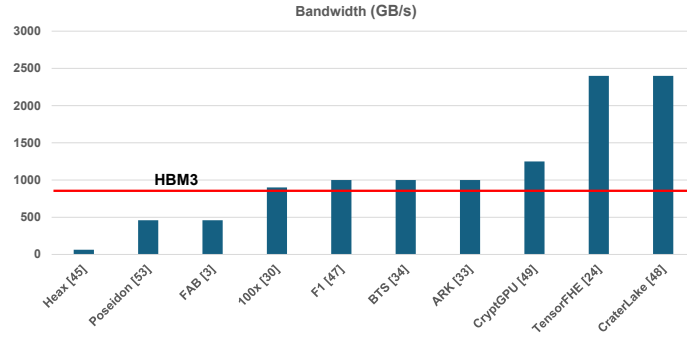


Fig. 3. Bandwidth demands of recent FHE accelerators.

reused across multiple butterfly operations [48]. This structured reuse significantly reduces both storage requirements and memory bandwidth, an aspect that is particularly advantageous for hardware acceleration. Exploiting twiddle-factor reuse enables efficient precomputation, caching, and parallel distribution of twiddle values, making Cooley–Tukey NTT especially well suited for high-throughput implementations in FHE accelerators.

In summary, NTT enables fast and exact polynomial multiplication in R_q , which is crucial for accelerating homomorphic encryption operations, including those in CKKS bootstrapping. Its integration with modular arithmetic ensures scalability and efficiency, making it a fundamental building block for practical lattice-based cryptography.

2.6 Existing FHE Accelerators

Since its inception in 2009 [28], FHE has achieved significant progress in reducing its initial computational overheads from 10^9 times slower than unencrypted computations. However, FHE is still $10,000\times$ to $100,000\times$ slower than traditional processing, posing a critical challenge for real-world applications and motivating the development of specialized hardware accelerators. GPUs, with their parallel processing capabilities, have demonstrated substantial speedups in FHE, achieving up to $257\times$ improvement over CPUs [32]. Open-source libraries like cuHE and cuFHE have further optimized FHE performance on GPUs [23, 55]. TensorFHE, for example, achieved a $1625.6\times$ speedup over CPU and a $2.9\times$ improvement over F1+, offering competitive performance to ASIC-based accelerators [26]. Despite these gains, GPUs are general-purpose and lack FHE-specific optimization, resulting in high power consumption and inefficiency for memory-intensive FHE operations [53]. FPGA-based accelerators provide greater adaptability, allowing for custom implementations of FHE operations like NTT. Solutions such as HEAX and Poseidon achieve significant performance gains, with Poseidon showing over a $1000\times$ speedup compared to GPUs [57]. Designs like FAB further optimize FHE-specific tasks through efficient resource management, demonstrating the potential of FPGA-based acceleration [3]. ASIC accelerators, designed explicitly for FHE schemes like CKKS and BFV, deliver even greater performance. Innovations in hardware bootstrapping and optimized data management, as seen in CraterLake [52] and ARK [35], enable extended computation depths and minimize bottlenecks, achieving orders-of-magnitude improvements over GPU-based approaches. However, these ASICs face challenges due to their large chip areas, high power demands, and substantial memory requirements, which can complicate real-world deployment.

2.7 Limitations of FHE Acceleration Trends

Data Inflation: Even with progress in compute acceleration, memory bandwidth continues to be a key bottleneck for FHE workloads [24]. In schemes such as CKKS, ciphertexts expand data size significantly compared to plaintexts, which in turn amplifies memory traffic. This inflation drives frequent memory transfers that cannot

be alleviated by compute-side improvements alone. *For example, one study shows that a chip with 40,960 modular multiplication units running at 2GHz alongside 3TB/s of HBM3 achieves all multiplications in 0.18ms, yet requires 2.1ms simply to load the data [35].* This more than tenfold delay illustrates how data movement, rather than arithmetic throughput, ultimately governs FHE scalability.

Memory Bandwidth: The irregular memory access patterns of FHE exacerbate bottlenecks, as accelerators must frequently access large twiddle-factor tables and intermediate results beyond on-chip cache. This forces repeated off-chip transfers, adding both latency and energy overhead. As shown in Fig. 3, FHE accelerators demand bandwidths from tens of GB/s to several TB/s. FPGA-based designs like HEAX [48] and Poseidon [57] require up to 64 GB/s, while ASICs such as BTS [36] and CraterLake [52] exceed 1 TB/s, with CraterLake reaching 2.4 TB/s. These demands surpass conventional memories such as HBM3, which peaks at 0.819 TB/s per stack. Techniques like 4-step FFT/NTT expose additional parallelism but increase storage for twiddle factors [6]. With $(n \log n)/2$ butterfly operations, hardware costs scale rapidly and static memory hierarchies fail to meet dynamic access needs [60]. This gap between demand and capability underscores the need for disruptive solutions such as photonic interconnects.

Dynamic Data Dependency: Key-switching operations add another layer of complexity. The decomposition parameter (d_{num}) directly influences both computation and memory demand, dictating the balance between NTT processing and basis conversion. Parallelization strategies such as coefficient-level parallelism (CLP) and residue-polynomial-level parallelism (rPLP) offer trade-offs: CLP increases latency due to global communication, while rPLP adds overhead from extra basis conversions. Although various optimizations exist, limited on-chip memory still forces accelerators to repeatedly access external memory for twiddle factors and intermediate data, which compounds latency and energy costs [61].

Bootstrapping Overhead: Bootstrapping remains one of the most prohibitive barriers to scalable FHE. In word-wise schemes, it involves multiple steps and consumes a significant portion of the available multiplicative depth, limiting the number of computations before refresh is required. Benchmark studies on encrypted neural networks have shown that bootstrapping can consume more than half of the available depth, which necessitates increasing polynomial degree n and ciphertext modulus Q to preserve security and functionality [52]. These increases enlarge ciphertexts and keys, inflating both memory footprint and computational load. Bootstrapping also requires numerous homomorphic permutations, especially in CoeffToSlot and SlotToCoeff conversions. In one implementation, processing a ciphertext that packs 4096 plaintexts required nearly 30 homomorphic rotations, underlining the significant cost of such operations [15]. Each permutation demands unique switching keys, which are often several times larger than ciphertexts, further straining memory and bandwidth. Together, these factors make bootstrapping a dominant contributor to latency, storage, and energy overhead in state-of-the-art accelerators.

As illustrated in Fig. 3, state-of-the-art accelerators already require data movement rates beyond what conventional electronic networks can deliver [24]. This imbalance underscores that alleviating the off-chip memory bottleneck is a prerequisite for practical FHE, particularly in workloads dominated by frequent bootstrapping. For domain-specific accelerators, enhancing interconnect bandwidth is therefore more critical than merely scaling compute resources, since without sufficient data transfer capability, additional computational logic cannot be effectively utilized.

3 Methodology

In this section we discuss the motivation behind *OptoLink*, provide an overview of the photonic interconnect implemented for *OptoLink*, the *OptoLink* network architecture, and implementation process.

3.1 Pathway to TB/s Bandwidth for FHE

To alleviate memory bandwidth bottlenecks, recent chiplet-based FHE accelerators have adopted high-bandwidth memory (HBM) solutions such as HBM3 [4]. A single HBM3 stack, with its 1024-bit interface, can deliver up to 0.819TB/s of bandwidth [31]. Meeting the multi-terabyte-per-second demands of modern FHE accelerators, as illustrated in Fig. 3, typically requires deploying several HBM3 stacks in parallel, aggregating thousands of data pins. Photonic interconnects provide a compelling alternative, as outlined below.

Ultra-High Bandwidth: Optical links inherently support massive bandwidth, making them highly attractive for FHE workloads. For instance, the proposed *OptoLink* design achieves a bandwidth of 0.8 TB/s using only 64 channels, which requires 16× fewer lanes than HBM3, while still maintaining scalability to meet the bandwidth demands of all accelerators illustrated in Fig. 3. This demonstrates *OptoLink*'s ability to outperform electronic interconnects while reducing the complexity and overhead of wide parallel I/O interfaces.

Flexible Routing: Beyond raw throughput, *OptoLink* also addresses the lack of flexibility in current FHE accelerators [60]. By supporting dynamic multiplexing across optical channels, it enables efficient routing of data between memory and compute units, improving utilization and reducing latency. This adaptability enables accelerators to efficiently handle diverse workloads, ranging from bandwidth-intensive NTT kernels to memory-heavy key-switching operations, without necessitating major architectural redesigns. The architectural details of *OptoLink* are described in Sec. 3.3, highlighting how its design enables ultra-fast, low-latency communication that exceeds traditional electrical approaches.

Support for Broadcast/Multicast: Photonic interconnects naturally enable one-to-many communication through wavelength-division multiplexing (WDM) [7]. This capability is particularly valuable for FHE kernels such as the NTT and key-switching, where identical twiddle factors or key polynomials must be distributed to multiple processing elements simultaneously. While electrical networks typically rely on complex crossbars or replication logic to support broadcast traffic [56], optical systems achieve this functionality with much lower overhead, thereby improving both scalability and efficiency.

Finally, Sec. 4 presents experimental validation of *OptoLink*'s performance, scalability, and robustness. Together, these results show that photonic interconnects can supply the bandwidth required for critical FHE kernels while preserving flexibility and efficiency, thereby advancing the scalability and real-world applicability of FHE in privacy-preserving computing.

3.2 Photonic Interconnects

Photonic interconnects represent a next-generation solution for on-chip and inter-chip communication, leveraging silicon photonics to replace conventional electrical signaling with high-speed, low-power optical links. As shown in Fig. 4, a typical architecture employs an external laser source that injects light into an on-chip waveguide through an optical coupler [42]. Once inside the chip, the optical signal is managed by micro-ring resonators (MRRs) [8], which serve as key components for modulation and filtering. Each MRR is thermally tuned with resistive heaters to maintain resonance stability despite fabrication or environmental variations [44].

The main operations in a photonic interconnect can be summarized as follows:

- **Transmission.** At the transmitter, electrical data is encoded onto the optical carrier using MRR modulators, with each channel assigned a distinct wavelength.
- **Reception.** On the receiver side, a second bank of MRRs selectively drops individual wavelengths to photodetectors (PDs), which convert the signals back into electrical form. Transimpedance amplifiers (TIAs) then restore the signal strength to ensure accurate recovery.
- **Wavelength-Division Multiplexing (WDM).** Multiple independent data streams propagate simultaneously within a single waveguide, each assigned a unique wavelength [59]. Dense WDM layouts have demonstrated up to 80 channels operating at 25 Gb/s each [34].

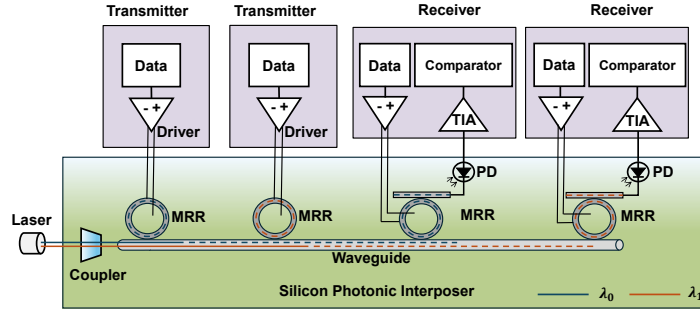


Fig. 4. Wavelength-division multiplexing (WDM) photonic interconnect. Electrical data is modulated onto distinct wavelengths (λ_0, λ_1) by micro-ring resonators (MRRs) at the transmitter, transmitted through a shared waveguide, and recovered by photodetectors (PDs) and TIAs at the receiver.

- **Space-Division Multiplexing (SDM).** WDM can be combined with SDM, transmitting parallel data streams across multiple waveguides [7]. This integration, together with advanced modulation formats, pushes the boundaries of throughput and efficiency.

By combining these multiplexing strategies, photonic interconnects deliver exceptional bandwidth efficiency and energy savings. Such characteristics make them especially well-suited for data-heavy workloads like FHE acceleration, where massive and frequent data exchanges between processing elements and memory subsystems demand communication technologies beyond the capabilities of electrical interconnects.

3.3 Single *OptoLink* Channel

Memory and NTT modules are connected within a single *OptoLink* channel using photonic interconnect as shown in Fig. 5. The architecture employs WDM to send multiple signals simultaneously through a single waveguide, with each signal assigned a distinct wavelength. Input data are stored in the memory, where they are converted to electric signals through digital-to-analog converters (DACs). Transmitter MRRs modulate certain wavelengths onto waveguides according to the converted electrical signals. On the receiving side, tuned MRRs filter out resonant wavelengths, directing each signal to a PD that converts it back into an electrical signal. This signal is then amplified by TIAs and processed by comparators to reconstruct the digital data. Once the data is processed by the NTT module, the results undergo the same demodulation and modulation for return to the memory controller to perform the next stage of the NTT computation. The same wavelengths can be utilized for input and output transmission because distinct waveguides will be utilized for input and output. This strategy of wavelength reuse also lowers the systems power consumption by lowering the number of wavelengths used. In CKKS bootstrapping, these NTT modules are the backbone of the CoeffToSlot/SlotToCoeff linear transforms, so *OptoLink* directly couples the memory to the processing engine to carry twiddle-factor and evaluation-key traffic efficiently.

3.4 Scalable *OptoLink* Network Architecture

The *OptoLink* setup in Fig. 6 employs five dedicated waveguides to connect a memory controller to four parallel NTT engines. *Waveguides 1* and *2* carry ciphertext coefficients, *Waveguides 3* and *4* deliver the twiddle factors, and *Waveguide 5* returns transformed outputs to memory. Two disjoint wavelength bands regulate traffic: $\lambda_1 - \lambda_{16}$ for inputs (coefficients and twiddles) and $\lambda_{17} - \lambda_{24}$ for outputs. Each waveguide carries one bit per optical channel; larger word widths are achieved by activating multiple channels in parallel using SDM. For example, a 128-channel *OptoLink* architecture deploys 128 waveguides in parallel.

Communication modes. The proposed interconnect is designed to support three communication modes: unicast, broadcast, and multicast. To illustrate these modes, we refer to the NTT algorithm adopted in the HEAX

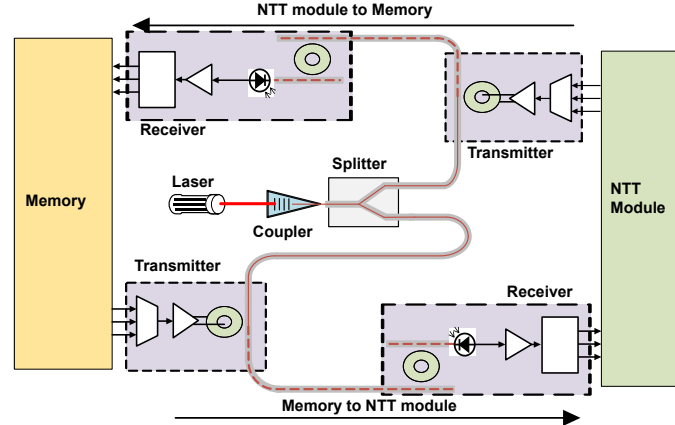


Fig. 5. Architecture of a single *OptoLink* channel using WDM to connect memory and an NTT module. Optical signals carry coefficients and twiddle factors to the NTT, while results are sent back to memory, enabling efficient bidirectional communication.

accelerator [48], where the polynomial under transformation consists of $n = 8$ coefficients (see Fig. 6). In this setting:

- *Unicast (stage inputs)*. At the input stage, each NTT core consumes a distinct, non-overlapping set of coefficients, so the memory controller sends them via unicast. Separate wavelengths on *Waveguide 1* (e.g., $\lambda_1 - \lambda_4$) are mapped one-to-one to the cores, and the receiver MRRs are biased on-resonance so the signal is fully dropped at the intended core (no through traffic), ensuring isolation.
- *Broadcast (stage-1 twiddles)*. In the first NTT stage $s = 0$, all butterfly operations reuse the same set of twiddle factors, as described in Sec. 2.5. To exploit this reuse, a single wavelength on *Waveguide 3* (e.g., λ_9) is power-split to every NTT by biasing receiver MRRs to partial resonance, dividing energy between drop and through ports. For larger n , the number of required broadcast wavelengths grows with the unique twiddle set, but remains independent of the number of cores, allowing scalable replication without duplication of data transfers.
- *Multicast (later-stage twiddles)*. Subsequent stages form core groups that share twiddles (e.g., cores 1–2 vs. 3–4). Multicast assigns one wavelength per group (e.g., λ_9 for cores 1–2 and λ_{10} for cores 3–4), combining targeted broadcast with unicast selectivity.

These quantitative allocation rules ensure that *OptoLink* scales seamlessly across NTT sizes and core counts. By assigning wavelengths based on data reuse patterns, the architecture minimizes redundant transmissions and avoids PE-memory conflicts, while maintaining predictable wavelength scaling across stages. Because receiver MRRs are dynamically tunable, *OptoLink* can switch among unicast, broadcast, and multicast at runtime, routing any wavelength to any subset of NTT cores without reconfiguring hardwired crossbars. This agility becomes more valuable as the polynomial size n grows and NTT dependency patterns become more irregular [60].

Communication modes for evaluation keys. Beyond NTT twiddle factors, a similar reuse pattern arises during key-switching in CKKS. As discussed in Sec. 2.3, evaluation keys are read-only operands that are repeatedly consumed across parallel engines, while ciphertext fragments are distinct and naturally partitioned. *OptoLink* leverages this asymmetric access pattern to efficiently distribute evaluation-key data using its broadcast and multicast capabilities.

- *Unicast (ciphertext fragments)*. Ciphertext limbs or digit blocks generated during key-switching are partitioned across processing engines, with each engine consuming a distinct subset. These private, non-reusable

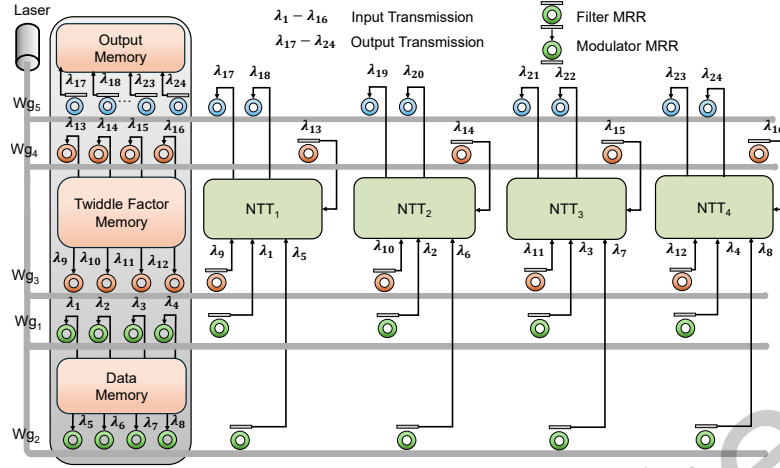


Fig. 6. Illustration of the *OptoLink* photonic interconnect for four NTT modules. Data memory, twiddle factor memory, and output memory communicate with NTT cores using wavelength-division multiplexing, where λ_1 – λ_{16} serve inputs and λ_{17} – λ_{24} carry outputs.

ciphertext fragments are therefore delivered via unicast, ensuring isolated, one-to-one transfers from the memory controller to each engine.

- **Broadcast (evaluation keys).** When all key-switching engines execute the same digit or RNS-limb step concurrently, they require the same evaluation-key block (e.g., a slice of evk_{mult} or $evk_{rot}^{(r)}$). In this case, the memory controller modulates the evaluation-key block onto a single wavelength and broadcasts it to all engines using the same power-split and partial-resonance MRR mechanism employed for stage-1 twiddle factors. This allows a single transmission to be optically replicated across all consumers, eliminating redundant key fetches.
- **Multicast (evaluation keys).** When engines are partitioned into groups processing different digit indices or RNS limbs (e.g., group A handling digit i and group B handling digit $i+1$), distinct evaluation-key slices are required by each group. *OptoLink* assigns different wavelengths to each group, multicasting the appropriate evaluation-key slice only to the engines that require it. This targeted replication follows the same principle as later-stage twiddle multicast, combining reuse within each group with selective delivery across groups.

Through concurrent WDM and SDM, *OptoLink* scales link bandwidth and fanout. Prior work indicates a single waveguide can host on the order of 64 WDM channels, providing substantial per-link throughput [45]. Moreover, MRR tunability lets a single transmitter address multiple receivers, lowering modulator count. The trade-off is power: lasers, MRR heaters, and photodetectors add to the energy budget. *OptoLink* balances these factors via wavelength reuse and mode selection (unicast/broadcast/multicast), delivering the throughput and routing flexibility required by high-throughput FHE pipelines while remaining practical across a range of application scales. Beyond coefficient and twiddle traffic, the same optical fabric can schedule burst transfers of evaluation keys for key-switching and bootstrapping, overlapping them with NTT execution. When evaluation-key digits are shared across multiple processing engines, multicast delivery amortizes bandwidth and reduces redundant off-chip memory accesses, enabling scalable and energy-efficient execution of high-throughput FHE pipelines.

Finally, a high-level overview of our 2.5D *OptoLink* architecture is shown in Fig. 7. Several processing units are integrated on a common silicon photonic interposer substrate, which provides low-latency optical connectivity between memory and the processing units, enabling scalable, high-bandwidth communication in a compact package.

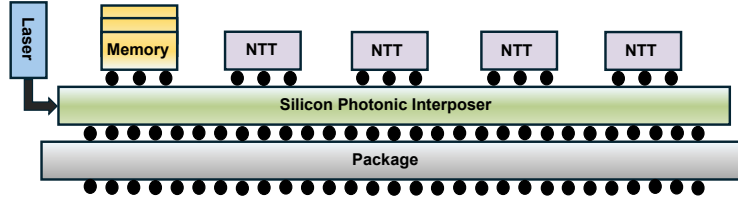


Fig. 7. Overview of proposed 2.5D *OptoLink* architecture.

3.5 Evaluation Metric (R)

To compare the electronic network and *OptoLink* on equal footing, we adopt an evaluation metric R that captures the joint impact of throughput, delay, and energy. Because photonic and electronic links differ substantially in raw bitrate and power, R provides a normalized efficiency measure across technologies.

$$R = \frac{\text{Bitrate}}{\text{Latency} \times \text{Power}} \quad (9)$$

Here, *Bitrate* denotes the transmitted data rate, *Latency* is the end-to-end transfer time, and *Power* is the total link power. Larger values of R indicate a more efficient interconnect, achieving high throughput while maintaining low delay and modest power consumption. Our goal is to maximize R for *OptoLink*, demonstrating superior efficiency relative to conventional electronic networks.

3.6 Bootstrapping Performance Evaluation

Evaluating the efficiency of bootstrapping is critical, as it remains the dominant bottleneck in CKKS-based FHE workloads. Following the methodology introduced by Han and Ki [30], we adopt a throughput-based performance metric that jointly reflects accuracy, ciphertext capacity, and runtime efficiency. The bootstrapping throughput is defined as

$$\text{Throughput} = \frac{n \cdot \log Q_1 \cdot bp}{brt}, \quad (10)$$

where n denotes the number of available plaintext slots, $\log Q_1$ represents the modulus size of the refreshed ciphertext (corresponding to the multiplicative depth it can support), bp is the plaintext bit precision, and brt is the bootstrapping runtime. Intuitively, this metric measures the amount of “useful work” performed per unit time by capturing how many slots can be refreshed at a given precision and depth within one bootstrapping cycle.

We use this throughput metric to assess the benefits of integrating *OptoLink* into FHE accelerators. In particular, our evaluation compares the throughput achieved with photonic interconnects against that of conventional electronic networks. By doing so, we aim to quantify how the increased bandwidth and reduced memory access latency of *OptoLink* translate into faster bootstrapping performance. This comparison highlights the effectiveness of *OptoLink* in mitigating memory bottlenecks that otherwise dominate bootstrapping cost in existing electronic network based accelerators.

3.7 Implementation Platform and Parameter Selection

To assess the effectiveness of the proposed *OptoLink* architecture, we implemented photonic interconnects linking processing modules with off-chip memory. This setup specifically targets the communication bottlenecks commonly encountered in FPGA-based FHE accelerators, such as HEAX [48]. The data movement patterns between memory and NTT in HEAX are particularly intensive, making them a suitable benchmark for exposing the shortcomings of traditional electronic interconnects. By contrast, the adoption of photonic links offers

a promising pathway to significantly improve bandwidth utilization and reduce latency in these demanding workloads.

For this study, we selected the CKKS security parameters listed in Table 2, which have been widely used in state-of-the-art accelerators such as HEAX, F1 and ARK. The polynomial degree N must be a power of two to enable efficient execution of the NTT. In practice, HEAX and F1 employ $N = 2^{13}$ and $N = 2^{14}$, respectively, reflecting platform-level trade-offs between security, performance, and hardware cost, while ARK adopts a larger polynomial degree of $N = 2^{16}$ to support higher security levels. Larger N values increase the ciphertext size but are necessary to support bootstrapping and to meet security requirements in realistic ML workloads. The ciphertext modulus $\log q$ determines the precision and multiplicative depth. For example, HEAX uses $\log q = 50$, whereas F1 and ARK adopt smaller moduli of $\log q = 32$, relying on deeper modulus chains or increased decomposition to meet computational requirements. The parameter L denotes the number of modulus levels in the CKKS modulus chain and determines the supported multiplicative depth. In our setup, HEAX uses $L = 30$, while F1 adopts a deeper chain with $L = 46$, enabling more complex bootstrapping and ML workloads at the cost of larger ciphertexts and increased memory traffic. The decomposition number ($dnum$) controls the number of digits in the RNS decomposition, where larger $dnum$ increases memory capacity and bandwidth demand but reduces per-digit complexity. Finally, the FFT iteration parameter ($fftIter$) affects the number of matrix–vector multiplications needed for linear transformations in bootstrapping.

To generate a realistic performance estimate for *OptoLink*, we employed Synopsys OptoCompiler to integrate key photonic device characteristics, such as detector responsivity, modulator insertion loss, and coupling efficiency (see Table 1). These parameters informed the calculation of the necessary laser power to ensure reliable data transmission in the presence of optical losses. In parallel, Synopsys Design Compiler was utilized to characterize the timing, power, and area of the electronic network. This methodology enabled benchmarking of *OptoLink* not only on primitive operations but also on full CKKS workloads, offering a detailed view of its capability to meet the bandwidth and memory demands of bootstrapping while maintaining efficiency.

4 Results and Analysis

This section presents the experimental evaluation of *OptoLink*, analyzing not only area, power, bandwidth, and latency, but also its impact on higher-level workloads such as CKKS bootstrapping and encrypted ML applications.

Table 1. Photonic Parameters used in *OptoLink*.

Component	Value
Laser Source	5 dB
Coupler	1 dB
Splitter	0.2 dB
Waveguide	1 dB/cm
Ring Drop	0.7 dB
Ring Through	0.01 dB
Photodetector	0.5 dB

Table 2. Security Parameters in FHE accelerators

Architecture	N	$\log q$	L	$dnum$	bit-precision	$fftIter$	Security Level
HEAX [48]	2^{13}	50	30	2	19	4	<128-bit
F1 [51]	2^{14}	32	46	2	24	4	<128-bit
ARK [35]	2^{16}	32	26	4	24	4	128-bit

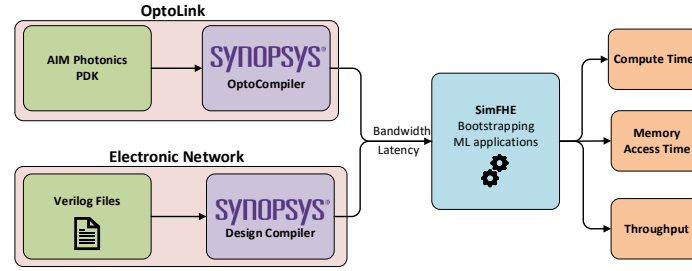


Fig. 8. End-to-end simulation flow for evaluating *OptoLink* and electronic interconnects. The photonic network is modeled using the AIM Photonics PDK within Synopsys OptoCompiler, while the electronic network is synthesized and analyzed using Synopsys Design Compiler. The resulting bandwidth and latency metrics are integrated into SimFHE, which simulates CKKS-based workloads such as bootstrapping and encrypted ML applications to estimate compute time, memory access time, and overall throughput.

Together, these results highlight the end-to-end advantages of *OptoLink* over traditional electronic networks in accelerating FHE.

4.1 Experimental Setup

To determine the area of the electrical network, a 32 nm technology library was employed, ensuring that our analysis reflects realistic design constraints and provides accurate estimates of physical footprint relative to *OptoLink*. In addition, we used Synopsys OptoCompiler and Synopsys Design Compiler to obtain latency and bandwidth characteristics of both the *OptoLink* and traditional electronic interconnects. These simulations were performed using the AIM Photonics PDK for photonic component modeling and Verilog-based synthesis for electronic designs, as summarized in Fig. 8. To complement this hardware-level evaluation, we relied on SimFHE, a custom simulator designed to model CKKS operations [2]. SimFHE tracks the execution of primitive operations and bootstrapping by accounting for both computational costs (in terms of modular multiplications and additions) and memory activity, including DRAM transfers for a given cache size. This explicit modeling of memory transfers is particularly important, as it directly captures the off-chip bandwidth bottlenecks that dominate bootstrapping performance in current FHE accelerators. For our study, we further extended SimFHE to incorporate the measured bandwidth and latency characteristics of photonic interconnects, enabling a direct comparison against traditional electronic networks. This extension allowed us to simulate CKKS operations under both interconnect paradigms, thereby providing a baseline characterization of bootstrapping performance and highlighting the raw benefits of integrating photonic links into the FHE acceleration pipeline.

For benchmarking, we adopted hardware configurations consistent with state-of-the-art FHE accelerators shown in Table 3. Specifically, the HEAX [48] setup operates at 0.3 GHz with 28.6 MB of on-chip memory, 256 processing elements, and 64 GBps memory bandwidth, while the F1 accelerator [51] runs at 1 GHz with 64 MB of on-chip memory, 18,432 processing elements, and 1000 GBps bandwidth. We also consider the ARK accelerator [35], configured at 1 GHz with 512 MB on-chip memory, 20,480 processing elements, and 1000 GBps bandwidth, to evaluate *OptoLink* under a high-parallelism, high-bandwidth ASIC setting. These configurations were used to evaluate the performance of CKKS bootstrapping and to assess end-to-end encrypted workloads, including logistic regression (LR) model training [29] and ResNet-20 inference [37].

4.2 Timing Analysis

To evaluate the timing performance of the *OptoLink* network, we conducted data transmission experiments using two optical channels. The simulation, performed in Synopsys OptoCompiler, included a single-channel setup as shown in Fig. 9. A pseudo-random bit sequence (PRBS) generator produced a 10 Gb/s data stream, requiring

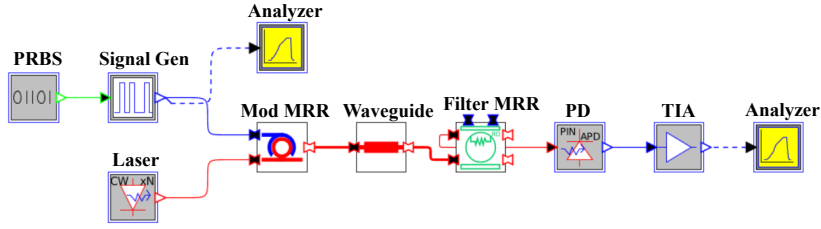


Fig. 9. Experimental setup for simulating a single *OptoLink* channel. A PRBS source drives the modulator MRR, transmitting optical signals over a $1000\text{ }\mu\text{m}$ waveguide at 1550 nm . The received signals are filtered, detected by a PD, amplified with a TIA, and analyzed for integrity.

6.4 ns to generate a 64-bit sequence. This sequence was modulated onto distinct wavelengths using MRRs, with 1550 nm assigned to channel 1 and 1551 nm to channel 2. At the receiver, the recovered signals were converted by TIAs, confirming reliable transmission and data integrity, as shown in Fig. 10(c–d). Transmission through a $1000\text{ }\mu\text{m}$ waveguide introduced only 10 ps delay, underscoring the ultra-low latency of optical propagation. In comparison, the corresponding electrical interconnect, modeled with Synopsys Design Compiler using the HEAX NTT design [48], required approximately 3.04 ns for the same transfer, highlighting the substantial latency advantage of *OptoLink*.

From this 10 ps delay, the data rate of a single *OptoLink* channel is 100 Gb/s (12.5 GB/s). Scaling to 128 parallel channels yields an aggregate throughput of 1.6 TB/s , sufficient for large-scale FHE workloads (Fig. 3). A 192-channel configuration achieves 2.4 TB/s , comparable to the NVIDIA A100 [21], while extrapolation to 1024 channels indicates a peak of 12.8 TB/s . By contrast, electronic networks in HEAX are limited to only 5.26 GB/s at 128-bit width and 42.1 GB/s even at 1024-bit width (Table 4). Achieving 1.6 TB/s with electrical connections would require an impractical 4864-bit bus, underscoring the scalability and efficiency advantage of photonic interconnects.

These high data rates are critical for FHE workloads, where large ciphertexts, twiddle factors, and evaluation keys must frequently be exchanged between memory and compute units. Such data often spans several gigabytes and dominates runtime in current accelerators [3, 52]. By enabling rapid transfer of these structures, *OptoLink* alleviates the memory bottleneck, accelerates bootstrapping and key-switching, and supports larger encrypted datasets, thereby improving both performance and scalability of privacy-preserving applications. To the best of our knowledge, this work is the first to introduce a photonic interconnect explicitly designed for FHE workloads. Prior photonic interconnect architectures, such as SwiftNoC [20] and FlexiShare [46], target on-chip router fabrics for many-core processors, where performance is evaluated in terms of network throughput under contention, packet injection rate, and arbitration efficiency. In contrast, *OptoLink* is architected as a dedicated memory–compute interconnect and is evaluated based on provisioned link bandwidth and end-to-end latency, which are the dominant metrics for bandwidth-intensive FHE data movement. As a result, the throughput figures reported in these NoC-oriented works are not directly comparable to the link-level bandwidth achieved by *OptoLink*.

Table 3. Hardware setup for FHE accelerators

Architecture	Processor Frequency (GHz)	On-chip Memory (MB)	Processing Elements	Bandwidth (GBps)
HEAX [48]	0.3	28.6	256	64
F1 [51]	1	64	18432	1000
ARK [35]	1	512	20480	1000

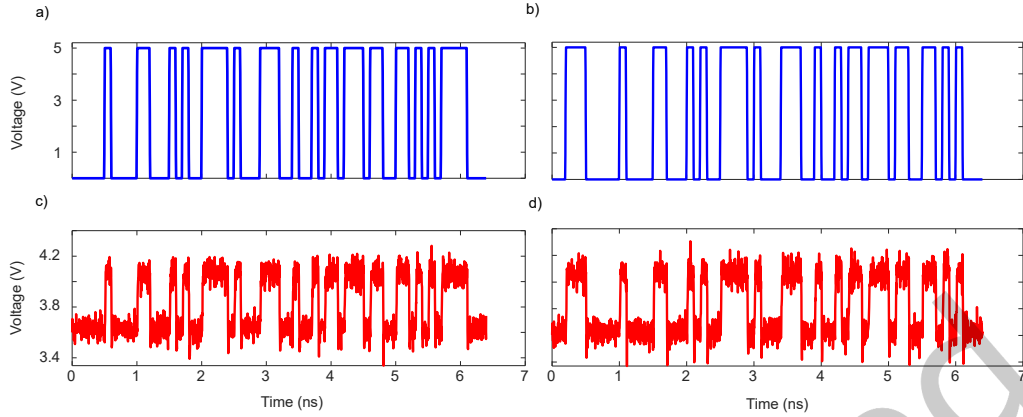


Fig. 10. (a–b) Input electrical waveforms driving the modulator MRRs for two distinct *OptoLink* channels at separate wavelengths. (c–d) Output signals after photodetection and TIA amplification, demonstrating reliable recovery of transmitted data.

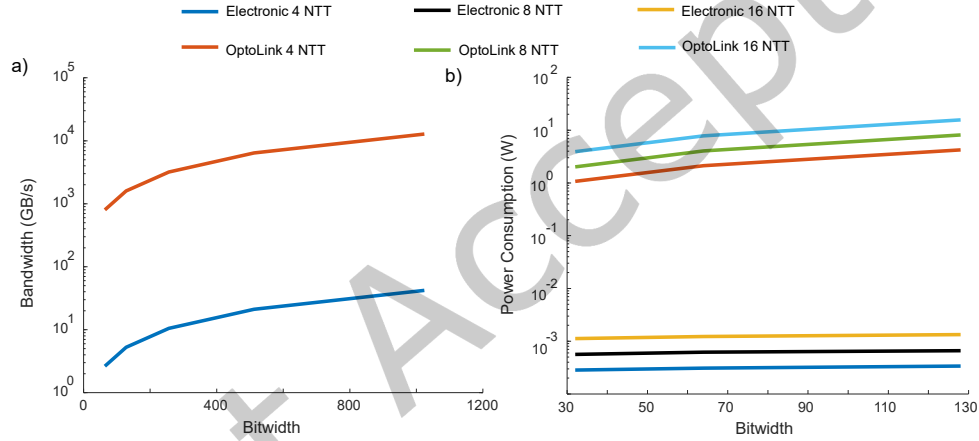


Fig. 11. Bandwidth and power comparison of electronic and *OptoLink* interconnects across varying bitwidths and NTT core counts: (a) bitwidth vs. bandwidth, (b) bitwidth vs. power consumption.

4.3 Power Consumption Analysis

The power profile of the proposed *OptoLink* system is primarily influenced by the laser source and the MRRs. The total consumption can be expressed as:

Table 4. Latency and bitrate comparison of electronic network and *OptoLink* for different bitwidths.

Bitwidth	Electronic Network		OptoLink	
	Latency	Bitrate	Latency	Bitrate
32	3.04ns	1.32GB/s	10ps	0.4TB/s
64	3.04ns	2.63GB/s	10ps	0.8TB/s
128	3.04ns	5.26GB/s	10ps	1.6TB/s

Table 5. Power consumption of electronic network and *OptoLink* across different bitwidths and NTT core counts.

Bitwidth	NTT Cores	Power Consumption	
		<i>Electronic Network</i> (μW)	<i>OptoLink</i> (W)
32	4	283.89	1.07
	8	562.44	2.12
	16	1121.9	4.23
64	4	308.18	2.02
	8	619.29	4.04
	16	1232.19	8.09
128	4	336.99	3.91
	8	661.74	7.82
	16	1332.31	15.63

Table 6. *OptoLink* power usage across NTT stages for 16 cores and multiple bitwidths. Earlier stages show lower power due to efficient broadcasting.

Bitwidth	NTT				
	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
32	3.38 W	3.48 W	3.68 W	4.09 W	4.91 W
64	6.75 W	6.96 W	7.37 W	8.19 W	9.82 W
128	13.5 W	13.92 W	14.7 W	16.37 W	19.65 W

$$P_{\text{total}} = P_{\text{laser}} + P_{\text{TX}} + P_{\text{RX}} \quad (11)$$

where P_{laser} denotes the power of the laser source, while P_{TX} and P_{RX} correspond to the power consumed by the transmitter and receiver sections, respectively. Based on prior evaluations, P_{TX} and P_{RX} are estimated to be 0.9 mW and 0.6 mW per channel [40].

Table 5 reports the power usage of *OptoLink* in comparison with an electronic interconnect across multiple configurations. For example, a system with 128 optical channels connected to 4 NTT modules consumes about 3.91 W. Each channel incorporates 24 transmitters and receivers, which dominate the total budget. Scaling the number of NTT cores increases power proportionally, reaching 7.82 W for 8 cores and 15.63 W for 16 cores. Likewise, higher operand bitwidths increase the required number of parallel channels via SDM, further raising power consumption. In comparison, the equivalent electronic interconnect, modeled using the HEAX architecture's NTT algorithm [48], requires significantly less power: 336.99, μW , 661.74, μW , and 1332.31, μW for 4, 8, and 16 NTT modules under a 128-bit configuration, respectively. This difference arises from the absence of optical devices such as lasers, modulators, and photodetectors in the electronic network.

Table 6 provides further insights for a 16-core setup, showing power across different NTT stages. Stage 1 broadcasting demonstrates an average power reduction of 31.2% compared to Stage 5. This efficiency results from optical broadcasting, where a single data stream is shared across multiple NTT modules, reducing the number of active wavelengths and consequently lowering the number of optical components required. The improvement becomes more pronounced for larger polynomial sizes, where a single broadcast supports more computation. These results underscore the potential of *OptoLink* as a highly energy-efficient solution for large-scale FHE acceleration.

4.4 Evaluation metric (R) Analysis

The evaluation metric defined in Eq. 9 clearly illustrates the performance disparity between *OptoLink* and traditional electronic network. For instance, in a setup with a 32-bit channel and 4 NTT cores, *OptoLink* achieves

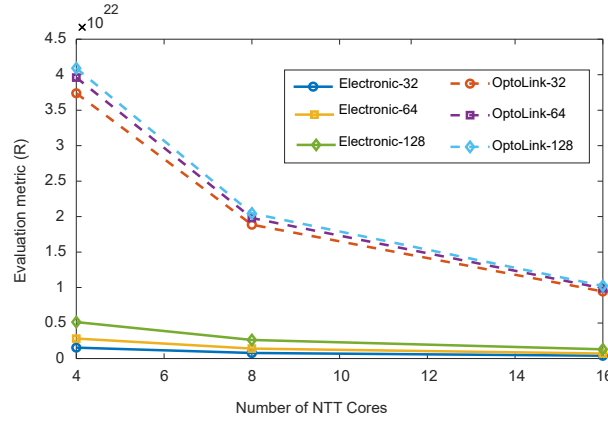


Fig. 12. Evaluation metric (R) comparison of electronic and *OptoLink* networks across different NTT core counts and bitwidths.

an R value of 3.79×10^{22} , compared to only 1.53×10^{21} for the electronic counterpart. This performance advantage is consistent across all tested configurations. As depicted in Fig. 12, *OptoLink* consistently achieves higher R values regardless of bitwidth or number of NTT cores, underscoring its scalability and efficiency.

The primary source of this improvement lies in the ultra-low latency and massive bandwidth provided by the photonic fabric, which reduces communication delays and alleviates memory bottlenecks. By efficiently sustaining data-heavy operations such as polynomial multiplications, bootstrapping, and key-switching, *OptoLink* makes it feasible to process large ciphertexts and evaluation keys that typically overwhelm electronic networks. In doing so, it establishes a strong foundation for practical and scalable FHE acceleration, demonstrating clear and consistent benefits over conventional interconnect technologies.

4.5 Area Analysis

The area footprint of the proposed *OptoLink* architecture was analyzed by contrasting the spatial requirements of conventional electronic network of HEAX architecture [48] with those of the photonic components integrated into *OptoLink*. For the electronic baseline, estimates were obtained using a 32, nm technology library to ensure realistic process-level accuracy. Under a 128-bit NTT configuration, the silicon area of electronic interconnects scales nearly linearly with the number of NTT units, occupying $3097.3\mu\text{m}^2$, $5741.2\mu\text{m}^2$, and $11861.9\mu\text{m}^2$ for 4, 8, and 16 units, respectively. By comparison, *OptoLink* incurs a larger area overhead due to the inclusion of photonic devices. Prior studies indicate that each photonic transmitter or receiver requires roughly 0.0096mm^2 per wavelength [54]. Additionally, MRRs, which are essential for wavelength-selective modulation, contribute further to the footprint. With a radius of approximately $5\mu\text{m}$, the aggregate area overhead from MRRs is estimated to be around 0.01mm^2 [38].

5 Case Studies

This section examines the benefits of *OptoLink* across two representative domains: CKKS bootstrapping and encrypted ML workloads. Together, these case studies illustrate how *OptoLink* enhance both core FHE primitives and end-to-end applications, reinforcing their role in enabling scalable privacy-preserving computation.

5.1 Bootstrapping Performance Analysis

To assess bootstrapping performance under different interconnect paradigms, we extend SimFHE to model both electronic and photonic networks. SimFHE estimates runtime by tracking the number of modular arithmetic operations and DRAM transfers for a given on-chip memory size as shown in Table 3. Compute time is derived

from the total operations, the processing frequency, and the level of parallelism available in each architecture (Table 3). The corresponding CKKS security parameters, which define polynomial degree, modulus size, and bit-precision, are summarized in Table 2.

Memory transfer time was computed as the ratio of effective data movement, including ciphertext limbs, twiddle factors, and auxiliary key material such as rotation, relinearization, and bootstrapping keys, to the available interconnect bandwidth. Results show that *OptoLink* dramatically reduces memory time compared to electronic networks. Specifically, in the HEAX accelerator, *OptoLink* achieves 0.96 ms versus 293.1 ms for electronics, representing more than a 300× improvement. In the F1 design, the reduction is more modest but still significant, with *OptoLink* lowering memory time from 18.774 ms to 11.734 ms, a 1.6× speedup. A similar trend is observed for the ARK architecture. As reported in Table 7, *OptoLink* lowers ARK’s memory transfer time from 62.232 ms with electronic interconnects to 38.895 ms, corresponding to a 1.6× reduction. These improvements can also be seen in Fig. 13(a), where *OptoLink* reduces memory latency to 0.0033× for HEAX and 0.63× for both F1 and ARK relative to the electronic network.

Throughput was evaluated using the bootstrapping throughput metric defined in Eq. 10. As summarized in Table 7, *OptoLink* achieves 1.38×10^6 bits/ms for HEAX, compared to only 1.26×10^5 bits/ms with the electronic network, yielding nearly an 11× improvement. For F1, throughput increases from 2.21×10^7 bits/ms with electronics to 3.54×10^7 bits/ms with *OptoLink*, providing a 1.6× gain. ARK also benefits substantially from *OptoLink*. Bootstrapping throughput improves from 1.05×10^7 bits/ms with electronic interconnects to 1.68×10^7 bits/ms using *OptoLink*, representing a 1.6× increase. These throughput advantages are further illustrated in Fig. 13(b), which highlights *OptoLink*’s superior performance across all architectures. It is worth noting, however, that in HEAX the throughput improvements are constrained by the compute core itself: the lower operating frequency and smaller number of processing units result in significantly longer compute times, making memory acceleration less dominant. In contrast, F1 and ARK’s higher degree of parallelism and faster clock allow *OptoLink*’s bandwidth advantages to translate more directly into throughput gains.

These findings underscore that memory transfer remains the dominant factor in bootstrapping performance. *OptoLink* significantly alleviates this bottleneck by delivering both lower latency and higher bandwidth. The benefits are especially pronounced in HEAX, where limited memory bandwidth severely constrains performance, while F1 and ARK, which is already equipped with HBM, still exhibits meaningful improvements. Overall, *OptoLink* demonstrates consistent gains in throughput and scalability, highlighting the necessity of high-bandwidth, low-latency photonic interconnects for enabling practical large-scale FHE acceleration.

5.2 ML Application Performance Analysis

To further assess the benefits of *OptoLink* in practical workloads, we evaluate encrypted LR model training and ResNet-20 inference using SimFHE. The HELR algorithm was implemented for LR [29], while leveraging the security and hardware parameters listed in Tables 2 and 3. For both LR and ResNet-20, compute time is primarily dictated by the accelerator’s frequency and number of processing elements, while memory time is determined by the volume of ciphertext and key data transferred relative to interconnect bandwidth.

Table 7. Bootstrapping performance comparison on different FHE accelerators.

Architecture	Function	Network	Compute Time (ms)	Memory Time (ms)	Throughput (bits/ms)
HEAX [48]	Bootstrapping	<i>OptoLink</i>	26.73	0.96	1.38×10^6
		Electronic	26.73	293.108	1.26×10^5
F1 [51]	Bootstrapping	<i>OptoLink</i>	2.247	11.734	3.54×10^7
		Electronic	2.247	18.774	2.21×10^7
ARK [35]	Bootstrapping	<i>OptoLink</i>	5.374	38.895	1.68×10^7
		Electronic	5.374	62.232	1.05×10^7

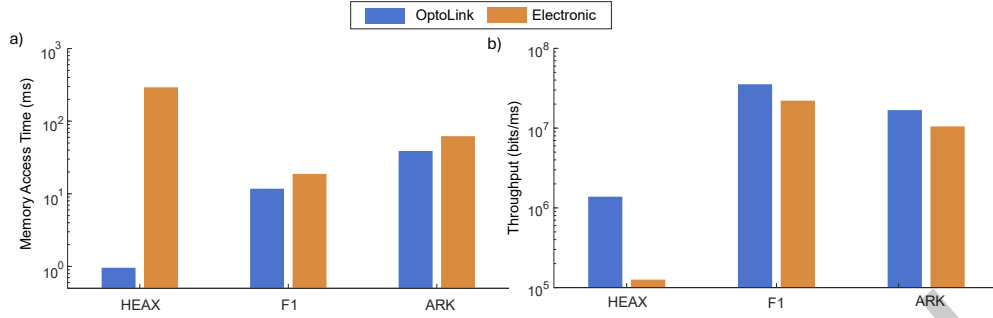


Fig. 13. Bootstrapping performance comparison of HEAX on a logarithmic scale. (a) Memory access time: *OptoLink* achieves lower memory latency than the electronic network, with improvements of 300× in HEAX and 1.6×. (b) Throughput: *OptoLink* sustains higher throughput compared to the electronic network, yielding 11× gains in HEAX and 1.6× in and ARK.

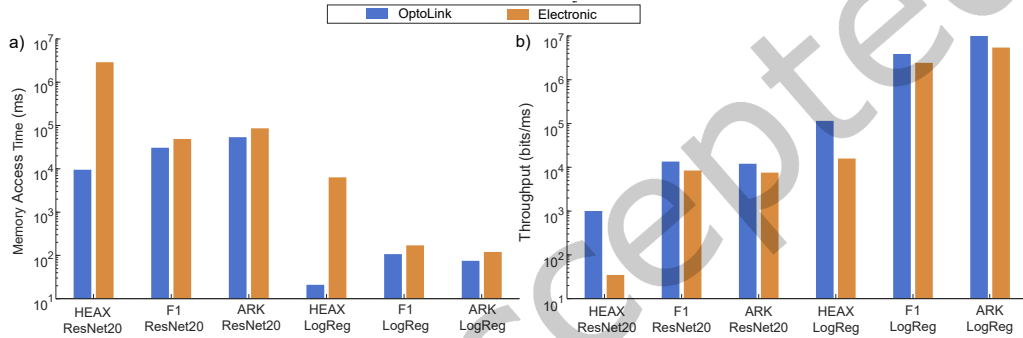


Fig. 14. Comparison of *OptoLink* and electronic interconnects for encrypted ML workloads on a logarithmic scale. (a) Memory access time across HEAX, F1 and ARK for ResNet-20 inference and logistic regression training. (b) Throughput for the same workloads. *OptoLink* achieves consistently lower memory latency and higher throughput compared to electronic interconnects.

Table 8 reports the absolute performance values, and Fig. 14 presents memory transfer time and throughput on a logarithmic scale. On HEAX, *OptoLink* reduces memory transfer time by more than 300× for both LR and ResNet-20. Performance comparison of LR training and ResNet-20 on FHE accelerators using electronic and *OptoLink* interconnects.

Architecture	Function	Network	Compute Time (ms)	Memory Time (ms)	Throughput (bits/ms)
LR	HEAX [48]	OptoLink	880.99	20.935	1.15×10^5
		Electronic	880.99	6367.92	1.59×10^4
	F1 [51]	OptoLink	16.208	106.906	3.88×10^6
		Electronic	16.208	171.05	2.43×10^6
	ARK [35]	OptoLink	9.758	75.137	1.09×10^7
		Electronic	9.758	120.219	5.44×10^6
ResNet-20	HEAX [48]	OptoLink	100791	9547.69	1003.77
		Electronic	100791	2904240	34.84
	F1 [51]	OptoLink	2638.74	30669.6	1.35×10^4
		Electronic	2638.74	49071.4	8.46×10^3
	ARK [35]	OptoLink	4317.16	53944.6	1.21×10^4
		Electronic	4317.16	86311.4	7.58×10^3

ResNet-20, cutting memory latency from 6367.9 ms to 20.9 ms for LR and from 2.90×10^6 ms to 9547.7 ms for ResNet-20. For F1, *OptoLink* reduces memory time by $1.6\times$ across both workloads (from 171.1 ms to 106.9 ms for LR and from 49071.4 ms to 30669.6 ms for ResNet-20). ARK exhibits similar trends. *OptoLink* reduces ARK's memory transfer time from 120.2 ms to 75.1 ms for LR and from 86.3 s to 53.9 s for ResNet-20, corresponding to approximately $1.6\times$ improvements in both workloads.

Throughput results mirror these improvements but remain partially limited by compute capacity, especially in HEAX where the low operating frequency and reduced parallelism dominate total runtime. For LR on HEAX, *OptoLink* increases throughput from 1.59×10^4 to 1.15×10^5 bits/ms, a $7.23\times$ gain. For ResNet-20, where memory traffic is much more dominant, throughput improves from 34.8 to 1003.8 bits/ms, which is a $28.8\times$ improvement. In contrast, F1 shows more modest gains, with throughput increasing $1.6\times$ in both LR and ResNet-20, reaching 3.88×10^6 bits/ms and 1.35×10^4 bits/ms, respectively. ARK also benefits from *OptoLink*, with throughput increasing from 5.44×10^6 to 1.09×10^7 bits/ms for LR and from 7.58×10^3 to 1.21×10^4 bits/ms for ResNet-20, as illustrated in Fig. 14(b).

Overall, these results highlight that the performance impact of *OptoLink* scales with the memory-boundedness of the workload. Architectures like HEAX, which suffer from severe bandwidth constraints, benefit most dramatically, while architectures like F1 and ARK with HBM-based memory already provisioned for high throughput still see consistent but smaller improvements. The trends illustrated in Fig. 14 confirm that *OptoLink* consistently reduces memory time and enhances throughput across both accelerators, thereby enabling more efficient execution of large-scale encrypted ML workloads.

6 Discussion

The results show that *OptoLink* provides significant advantages for memory-bound FHE workloads by delivering ultra-low latency and TBps bandwidth, far exceeding the limits of electronic network. These gains directly translate into reduced memory transfer time and higher throughput, with improvements exceeding $300\times$ in HEAX and $1.6\times$ in F1 and ARK. While the benefits are most pronounced in bandwidth-constrained designs, even accelerators with high-bandwidth HBM show consistent improvements. The primary trade-offs of *OptoLink* lie in its higher power usage, dominated by lasers and photonic devices, and larger area overhead compared to electronic interconnects. However, features such as optical broadcasting mitigate energy costs, and advances in silicon photonics integration continue to reduce area. Overall, *OptoLink* complements existing architectures by alleviating the memory bottleneck that dominates FHE, enabling faster bootstrapping and encrypted ML workloads, and pointing toward scalable next-generation privacy-preserving accelerators.

7 Conclusion

This work introduced *OptoLink*, a photonic interconnect architecture designed to address the memory bandwidth bottlenecks that limit scalability in state-of-the-art FHE accelerators. By leveraging WDM and SDM-enabled optical channels, *OptoLink* provides ultra-high throughput, flexible routing, and efficient broadcast capabilities, achieving terabyte-per-second bandwidths with significantly lower latency than conventional electronic networks. Through detailed simulation and system-level evaluation, we demonstrated that *OptoLink* reduces memory transfer times by up to $300\times$ in bandwidth-constrained FPGA accelerators such as HEAX and still delivers consistent $1.6\times$ gains in advanced HBM-equipped designs like F1 and ARK. These improvements translate into faster bootstrapping, higher throughput in encrypted ML workloads, and enhanced scalability for privacy-preserving computation.

While photonic integration introduces higher power and area costs compared to electrical links, our analysis shows that optical broadcasting and wavelength reuse mitigate some of these overheads, and ongoing advances in silicon photonics are expected to further reduce integration costs. As a next step, we plan to pursue hardware prototyping of the *OptoLink* interconnect through fabrication of silicon photonic devices and their co-integration with FPGA or ASIC platforms. Future work will also focus on developing power-optimized designs, particularly

by exploring laser sharing, adaptive wavelength allocation, and energy-efficient driver circuits, to make *OptoLink* not only high-bandwidth but also power-competitive with next-generation interconnect technologies.

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